

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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MODEL PRACTICAL EXAMINATION – SET 2

*Answer all questions. Each question carries equal marks (20).
marks: 100*

Total

1. Create a simple cloud software application for Car Booking Reservation System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Search for Cabs, From, To, Rental, Out Station, Package Type, Hours/Days, etc.
2. Create a simple cloud software application for Hospital Information System for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS. Include the fields of Patient Name, Age, Address, Phone, Email, Attender Name, Attender Phone, Doctor Name, Diseases, etc.
3. Change the hardware compatibility of a VM (either by clone/create a new one) which is already created and configured, and run simple UNIX commands.
4. Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

1. Create a simple cloud software application for Car Booking Reservation System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Search for Cabs, From, To, Rental, Out Station, Package Type, Hours/Days, etc.

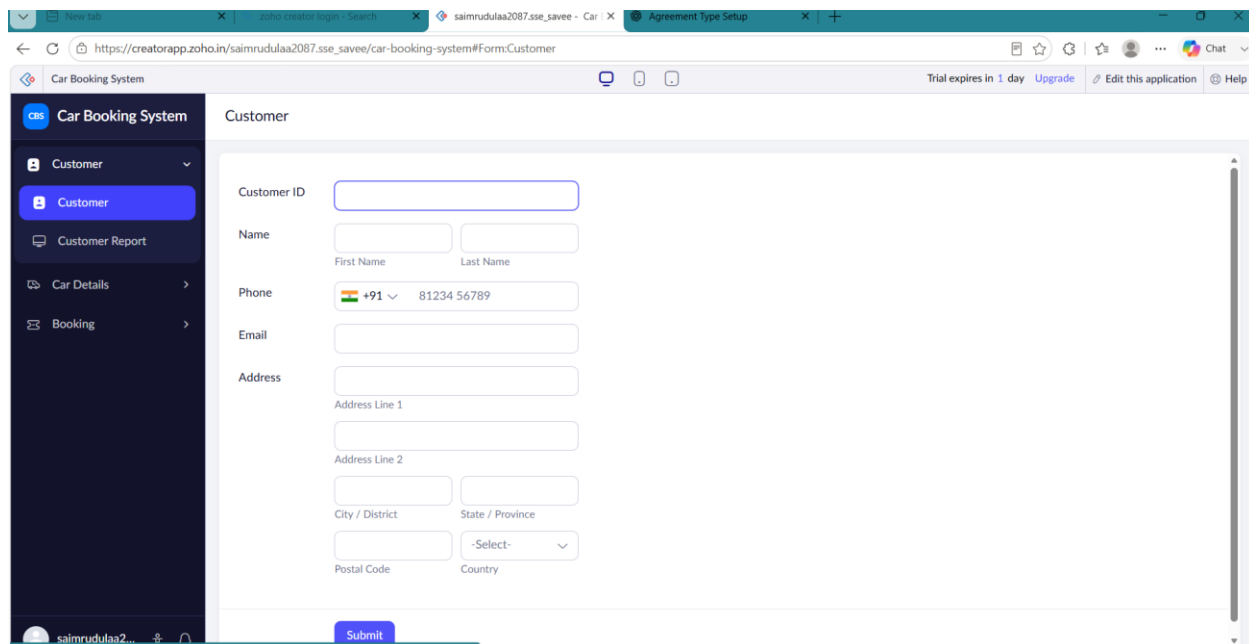
Aim:

To develop a cloud-based Car Booking Reservation System for booking vehicles and managing customer reservations.

Procedure:

1. Create a new application.
2. Create Customer, Car Details, and Booking forms.
3. Add booking information including pickup location, destination, package type, and rental duration.
4. Link Customer and Car forms using Lookup fields.
5. Generate booking reports.
6. Test booking functionality.
7. Publish the application.

Implementation:



The screenshot displays a web browser window with multiple tabs. The active tab shows a Zoho Creator application titled "Car Booking System". The URL in the address bar is https://creatorapp.zoho.in/saimrudulaa2087.sse_savee/car-booking-system#Form:Customer. The application interface features a dark sidebar on the left with a menu containing "Customer", "Customer Report", "Car Details", and "Booking". The main content area is titled "Customer" and contains a form with the following fields: "Customer ID" (a single-line text input), "Name" (split into "First Name" and "Last Name" inputs), "Phone" (with a dropdown for country code set to "+91" and a text input for the number "81234 56789"), "Email" (a single-line text input), "Address" (split into "Address Line 1" and "Address Line 2" inputs), "City / District" and "State / Province" (two separate inputs), "Postal Code" (an input with a "Country" dropdown menu set to "-Select-"), and a "Submit" button at the bottom right. The top right of the application shows a trial status: "Trial expires in 1 day" with links for "Upgrade", "Edit this application", and "Help".

Car Booking System

Customer

Car Details

Car Details

Car Details Report

Booking

saimrudulaa2...

Car Details

Car ID

Car Name

First Name

Last Name

Vehicle Number

Car Type

-Select-

Seating Capacity

#####

Ac/Non AC

-Select-

Price

##,##,###

Driver Name

First Name

Last Name

Status

-Select-

Submit

30°C

Partly cloudy

Search

ENG

IN

10:43 PM

7/25/2026

Car Booking System

Customer

Car Details

Booking

Booking Report

saimrudulaa2...

Booking

Booking ID

Address Line 1

Address Line 2

City / District

State / Province

Postal Code

Country

Customer

-Select-

Car Details

-Select-

From

Address Line 1

To

Address Line 1

Hours

#####

30°C

Partly cloudy

Search

ENG

IN

10:43 PM

7/25/2026

Car Booking System

CBS

Car Booking System

Customer

Customer

Customer Report

Car Details

Booking

Customer Report

Name

Phone

Customer ID

Email

Address

Mrudula

+917801082407

1001

mrudula@gmail.com

Chennai, TamilNadu, India

Showing 1 of 1

Car Booking System

CBS

Car Booking System

Customer

Car Details

Car Details Report

Booking

Car Details Report

Car ID

Car Name

Vehicle Number

Car Type

Seating Capacity

Ac/Non AC

Price

101

Maruthi

TN23SM2006

Petrol

7

AC

₹ 5,00,000.00

Showing 1 of 1

Car Booking System

CBS

Car Booking System

Customer

Car Details

Booking

Booking Report

Booking Report

Booking ID

Customer

Car Details

From

To

Hours

Days

101

Mrudula

Maruthi

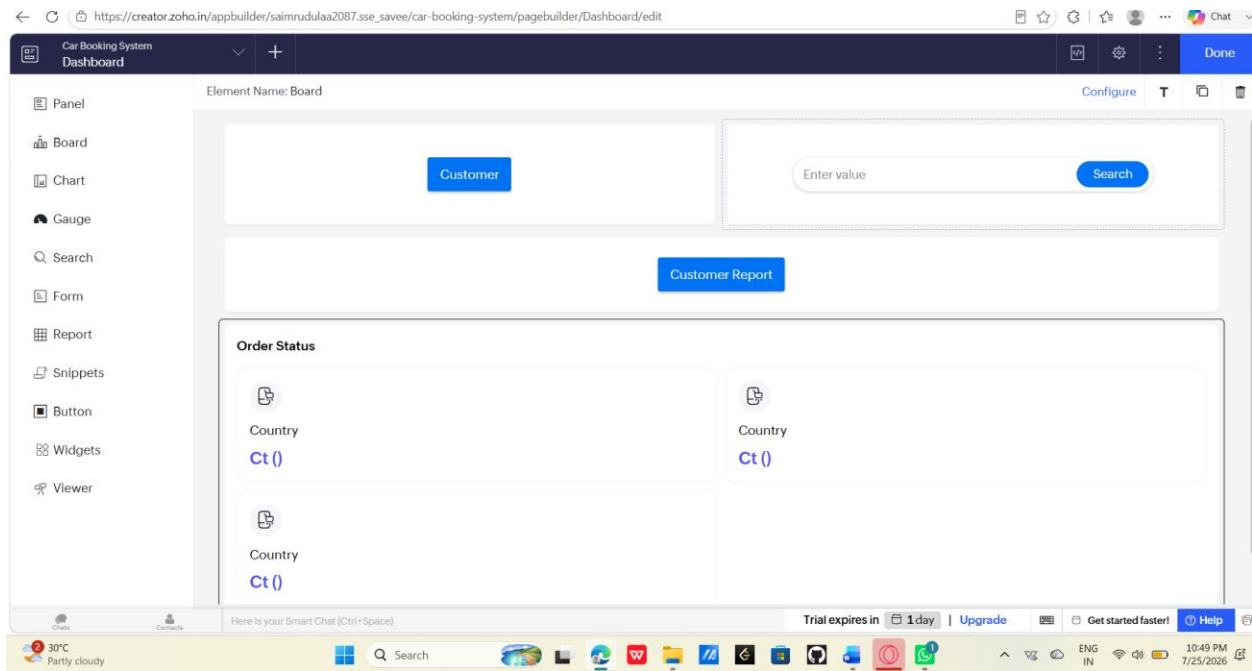
Chennai

Guntur

7

0

Showing 1 of 1



Result:

A cloud-based Car Booking Reservation System was successfully developed. The application manages vehicle reservations, rental packages, customer details, and booking reports effectively.

2. Create a simple cloud software application for Hospital Information System for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS. Include the fields of Patient Name, Age, Address, Phone, Email, Attender Name, Attender Phone, Doctor Name, Diseases, etc.

Aim:

To develop a cloud-based Hospital Information System for managing patient registration and hospital records.

Procedure:

1. Create Patient, Doctor, and Appointment forms.
2. Enter patient and doctor details.
3. Connect Patient and Doctor forms using Lookup fields.
4. Record consultation details.
5. Generate patient reports.
6. Test the application.
7. Publish the application.

Implementation:

The screenshot displays a web application for a hospital information system. The main form is titled 'Patient' and contains the following fields:

- Patient ID: A single-line text input field.
- Patient Name: Two text input fields labeled 'First Name' and 'Last Name'.
- Age: A text input field with a placeholder '#####'.
- Gender: A dropdown menu with the option '-Select-'.
- Address: Multiple input fields including 'Address Line 1', 'Address Line 2', 'City / District', 'State / Province', 'Postal Code', and 'Country'.
- Phone: A text input field with a country code dropdown set to '+91' and the number '81234 56789'.

The application interface includes a sidebar menu on the left with options: Hospital Details, Patient, Patient Report, Doctor, and Appointment. The top of the screen shows a trial expiration notice: 'Trial expires in 1 day' with an 'Upgrade' button and an 'Edit this application' link.

HPS Hospital information system for SIMAT...

Hospital Details

Patient

Patient Report

Doctor

Appointment

Trial expires in 1 day Upgrade Edit this application Help

Patient Report

Q + ...

☐ Patient ID	Patient Name	Age	Gender	Address	Phone	Email
24	sai mrudula	13	Female	guntur	+917546897665	sai12@gmail.com

Showing 1 of 1

Hospital information system for SIMATS Hospital

Trial expires in 1 day Upgrade Edit this application Help

Hospital information system for SIMAT...

Hospital Details

Patient

Doctor

Doctor

Doctor Report

Appointment

saimrudulaa2...

Doctor Report

Doctor ID Doctor Name Department Specialization Phone

678

tharun sai

cse

dentist

+919358780801

Showing 1 of 1

Hospital information system for SIMATS Hospital

Trial expires in 1 day Upgrade Edit this application Help

Hospital information system for SIMAT...

Hospital Details

Patient

Doctor

Appointment

Appointment Report

saimrudulaa2...

Appointment

Patient

Doctor

Disease

Attender Name

Attender Phone

Visit Date

Status

Submit

Hospital information system for SIMAT...

Hospital Details
Patient
Doctor
Appointment
Appointment
Appointment Report

Trial expires in 1 day
Upgrade
Edit this application
Help

Appointment Report

	Patient	Doctor	Disease	Attender Name	Attender Phone	Visit Date	Status
	24	678	hair loss	sai	+91823173345	26-Jul-2026	Not Cured

Showing 1 of 1

Result:

A cloud-based Hospital Information System was successfully developed. The application manages patient records, doctor consultations, appointments, and hospital reports efficiently.

3. Change the hardware compatibility of a VM (either by clone/create a new one) which is already created and configured, and run simple UNIX commands.

Changing Hardware Compatibility of a Virtual Machine

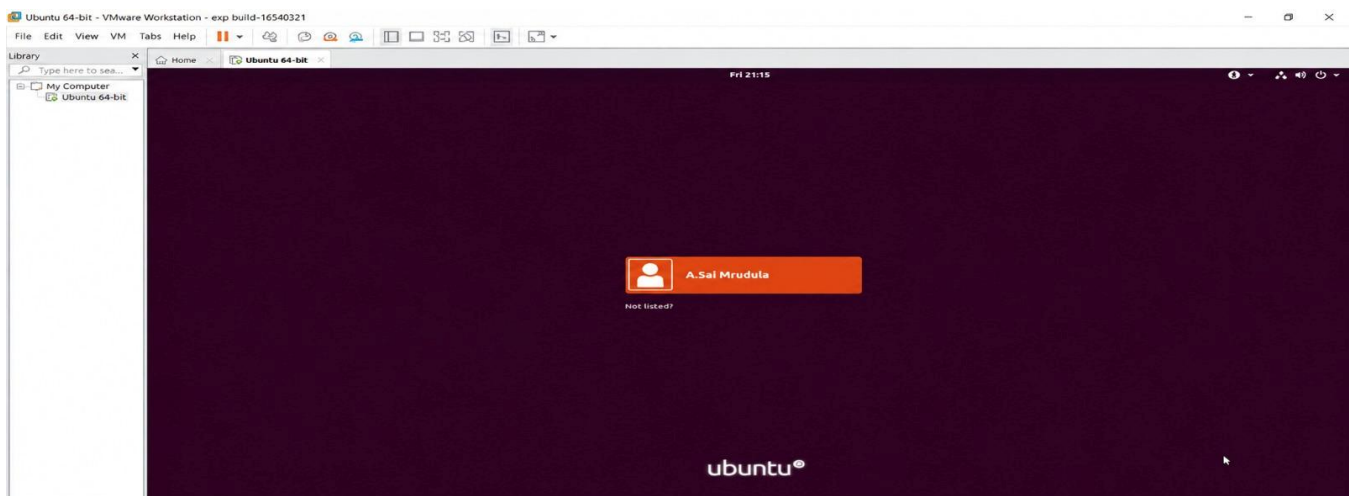
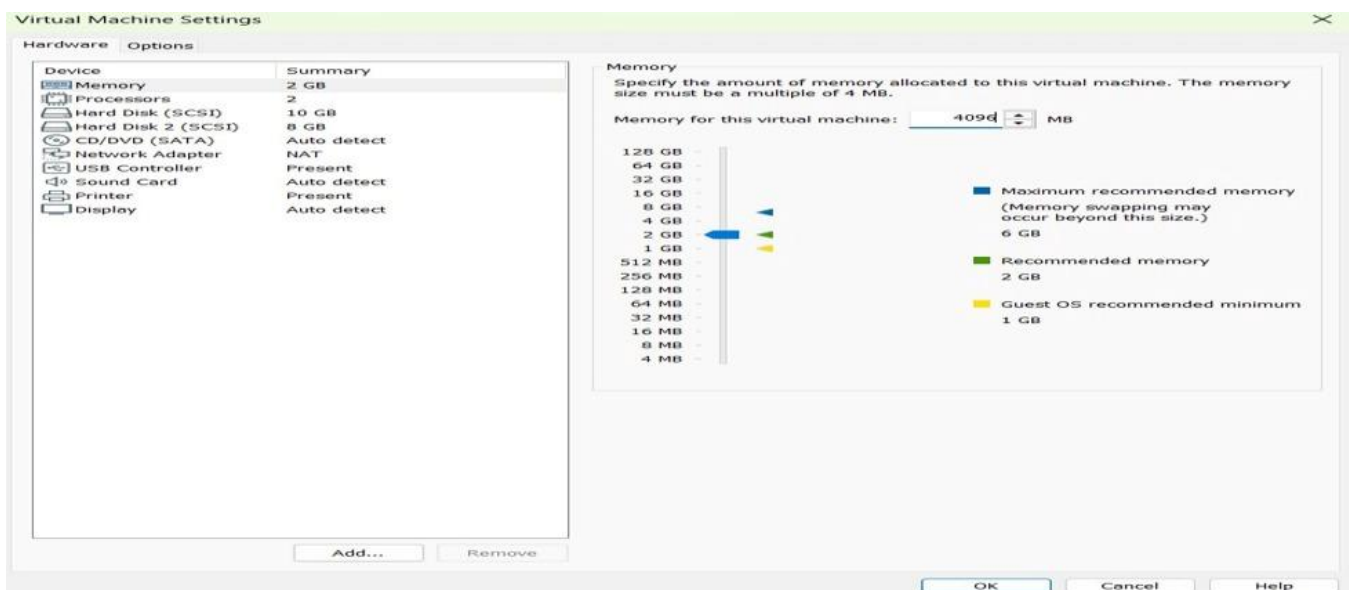
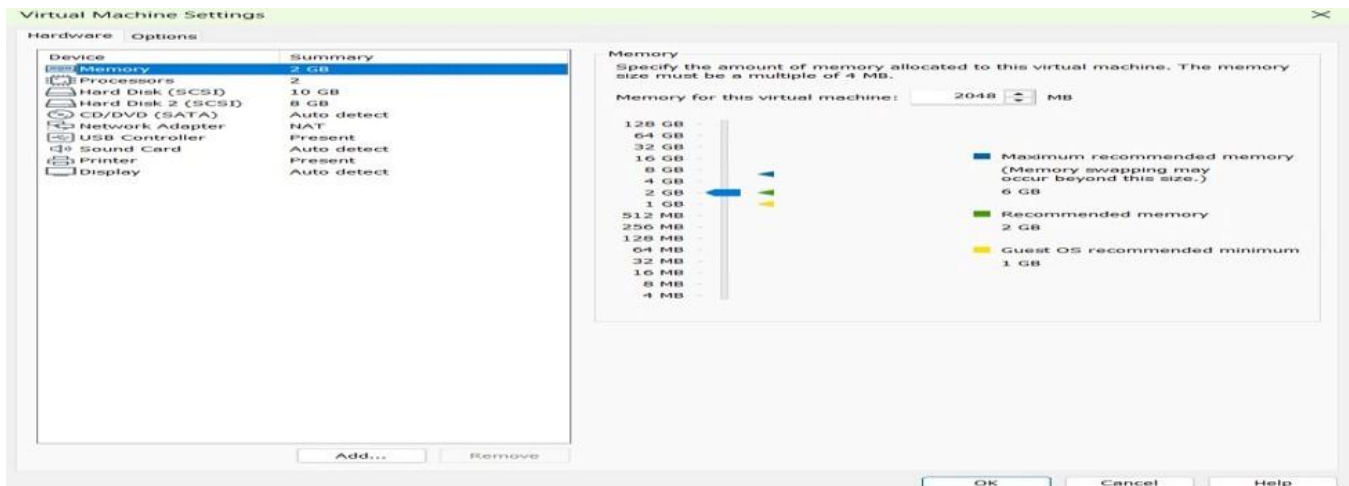
Aim

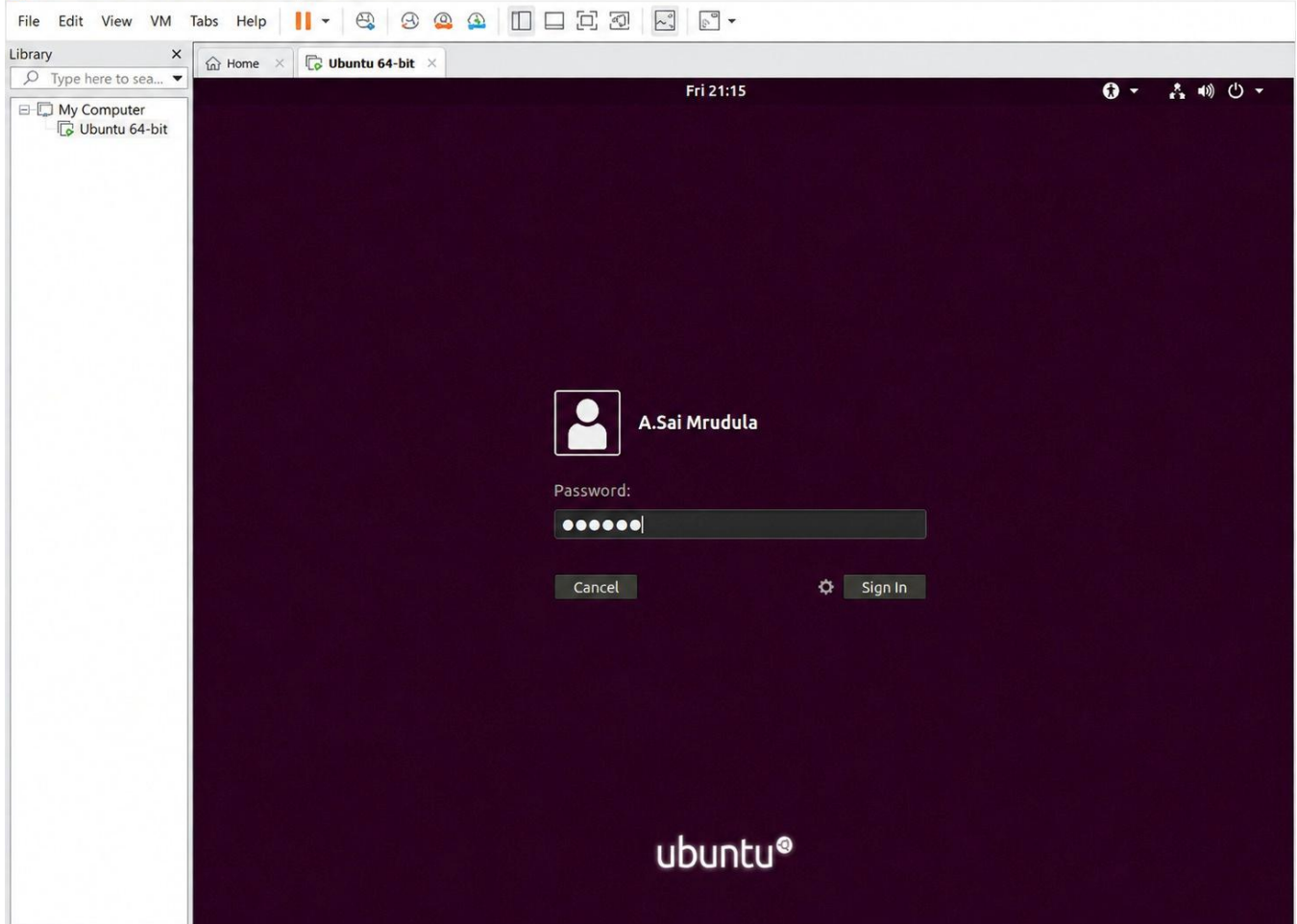
To change the hardware compatibility of an already created Virtual Machine by cloning or creating a new VM and execute simple UNIX commands.

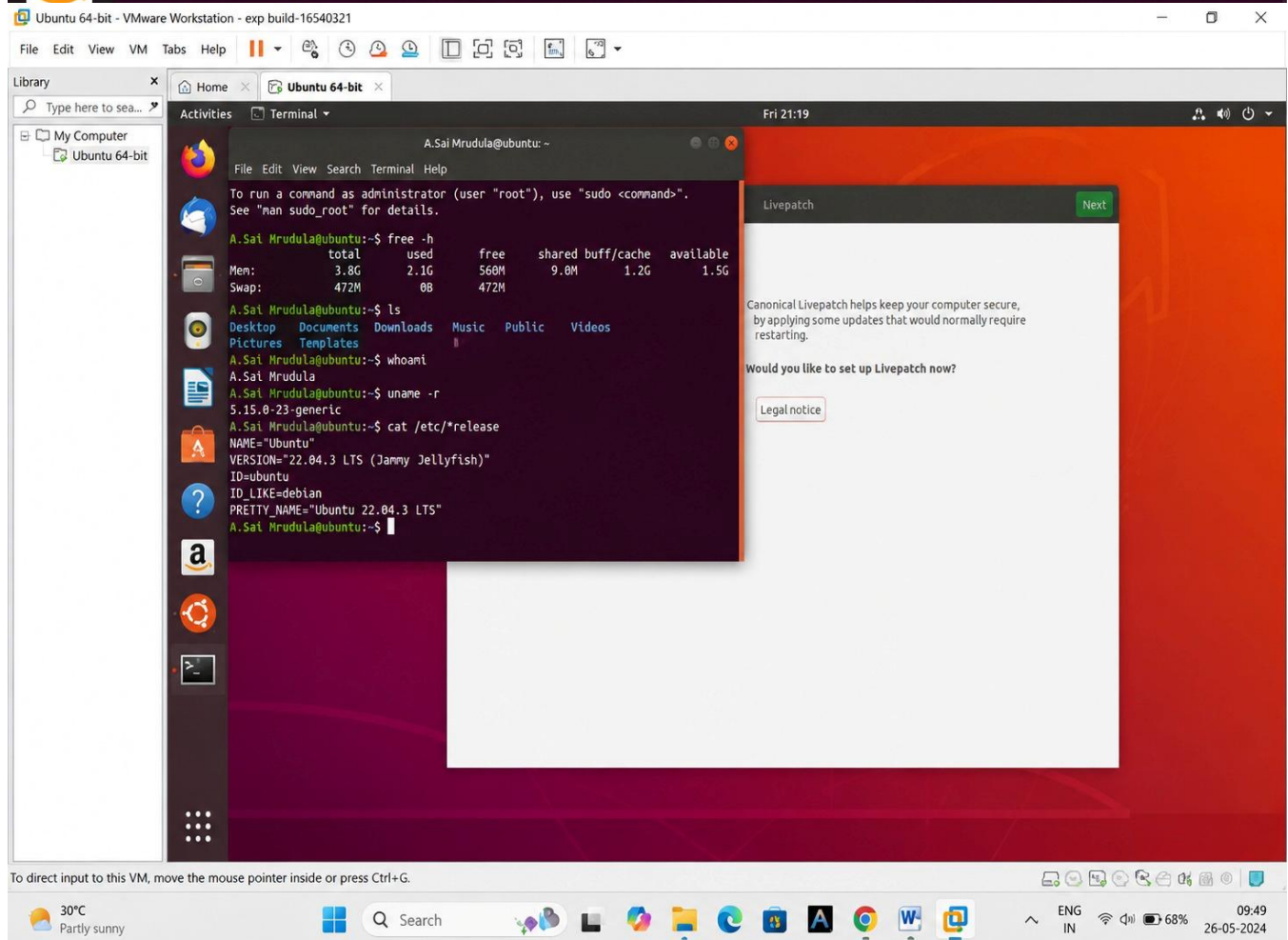
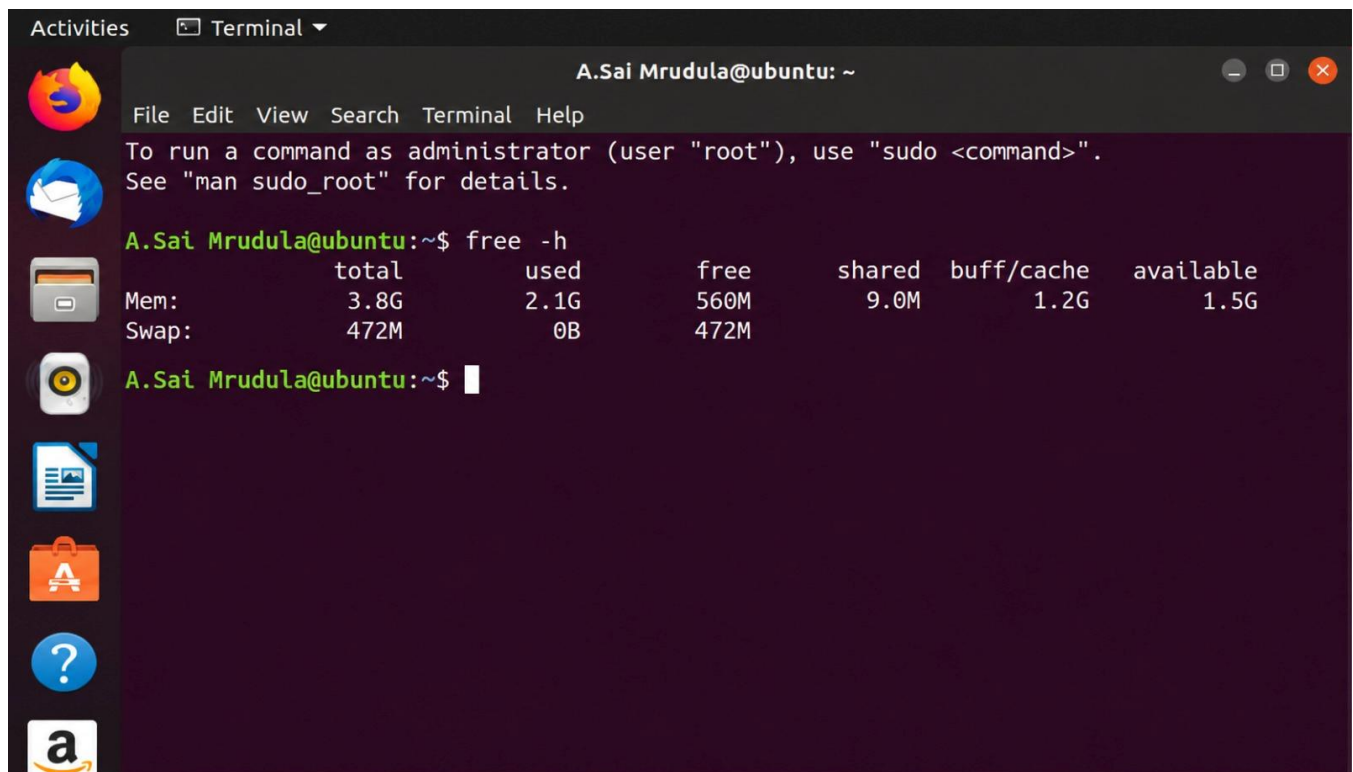
Procedure

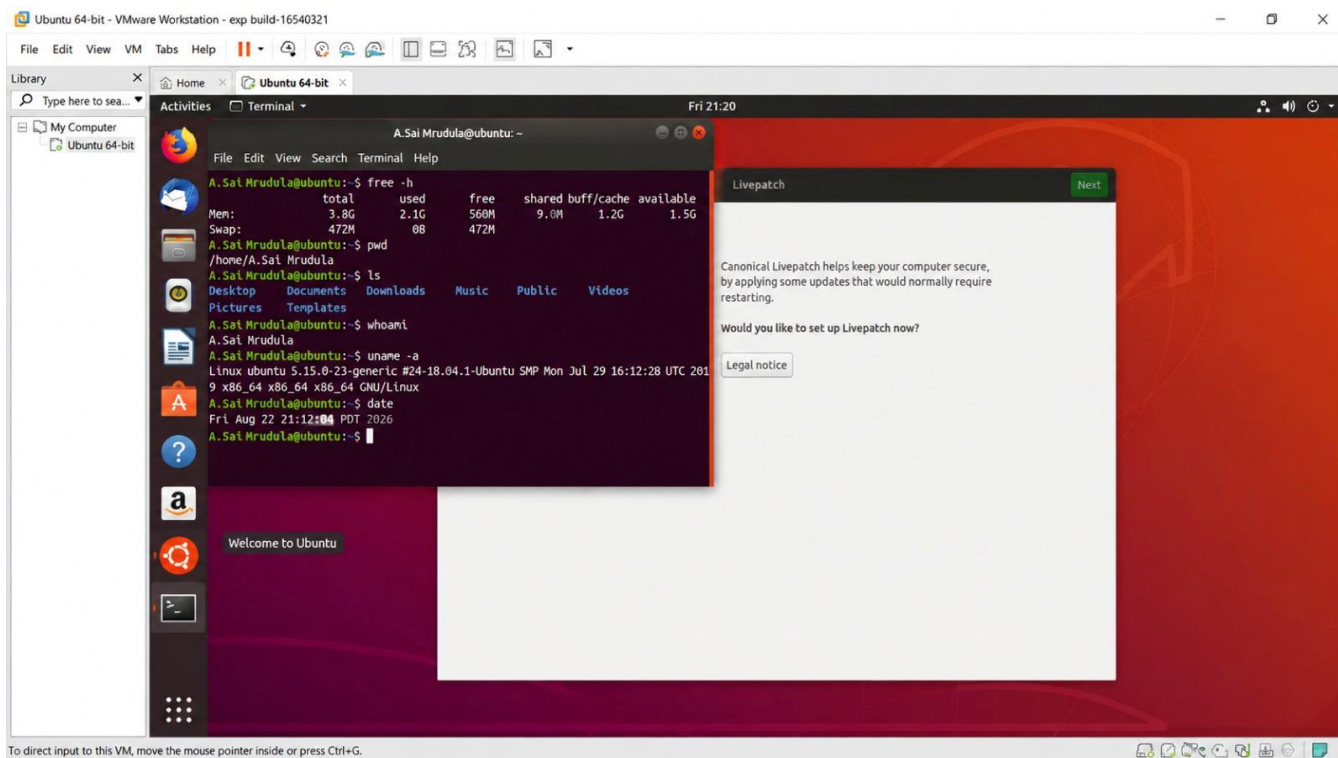
1. Open the Type-2 virtualization software such as VMware Workstation or VirtualBox.
2. Select an already created and configured Virtual Machine.
3. Take a snapshot or create a clone of the existing VM for safety.
4. Open the VM settings and modify the available hardware configuration.
5. Change the required hardware compatibility or virtual hardware settings.
6. Save the modified configuration.
7. Start the cloned or modified Virtual Machine.
8. Log in to the UNIX/Linux operating system.
9. Open the Terminal.
10. Execute simple UNIX commands such as:
 - pwd – displays the current directory.
 - ls – lists files and directories.
 - cd – changes the directory.
 - mkdir test – creates a directory.
 - date – displays the current date and time.
 - whoami – displays the current user.
11. Verify that the VM works correctly after changing the hardware compatibility.

Implementation









Result

The existing Virtual Machine was successfully cloned/modified with changed hardware compatibility. The VM started successfully, and basic UNIX commands were executed successfully.

4. Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

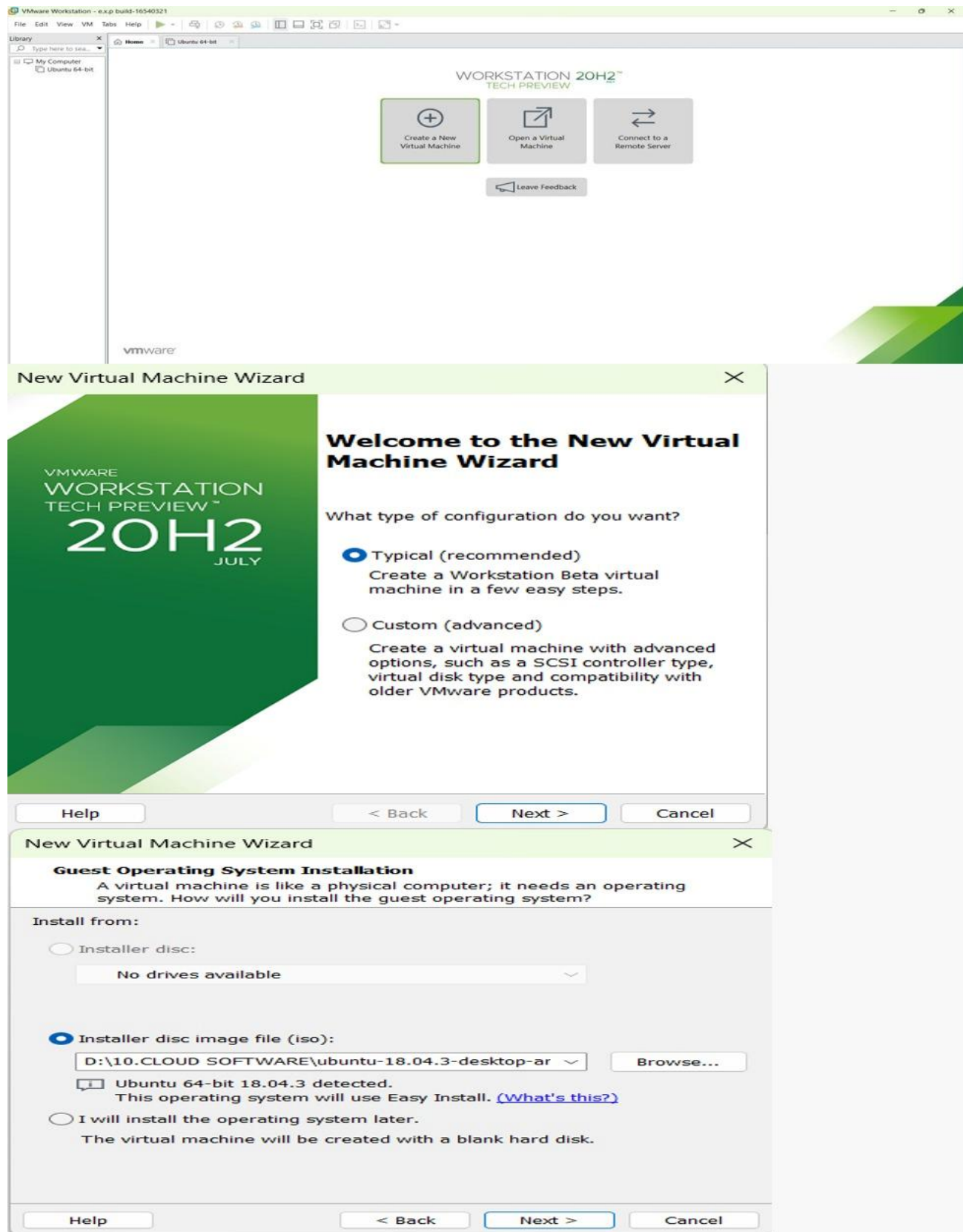
Aim

To create and configure a Virtual Machine with **1 CPU, 2 GB RAM, and 15 GB storage** using a Type-2 virtualization software.

Procedure

1. Install a Type-2 hypervisor such as VMware Workstation or Oracle VirtualBox on the host computer.
2. Open the virtualization software.
3. Select Create New Virtual Machine.
4. Enter a suitable name for the Virtual Machine.
5. Select the required guest operating system and its ISO image.
6. Configure the processor with 1 CPU/core.
7. Allocate 2 GB (2048 MB) RAM to the VM.
8. Create a new virtual hard disk with 15 GB storage.
9. Select the required virtual disk format and allocate the storage.
10. Review the configuration and create the VM.
11. Start the VM and install the selected operating system.
12. After installation, boot the VM and verify the CPU, RAM, and storage configuration.

Implementation



New Virtual Machine Wizard

Easy Install Information

This is used to install Ubuntu 64-bit.

Personalize Linux

Full name:

A.Sai Mrudula

User name:

asmrudula

Password:

••••••••

Confirm:

••••••••

Help

< Back

Next >

Cancel

New Virtual Machine Wizard

Name the Virtual Machine

What name would you like to use for this virtual machine?

Virtual machine name:

Ubuntu-15GB

Location:

C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-15GB

Browse...

The default location can be changed at Edit > Preferences.

< Back

Next >

Cancel

New Virtual Machine Wizard

Specify Disk Capacity

How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

15

Recommended size for Ubuntu 64-bit: 20 GB

☒ Store virtual disk as a single file

☐ Split virtual disk into multiple files

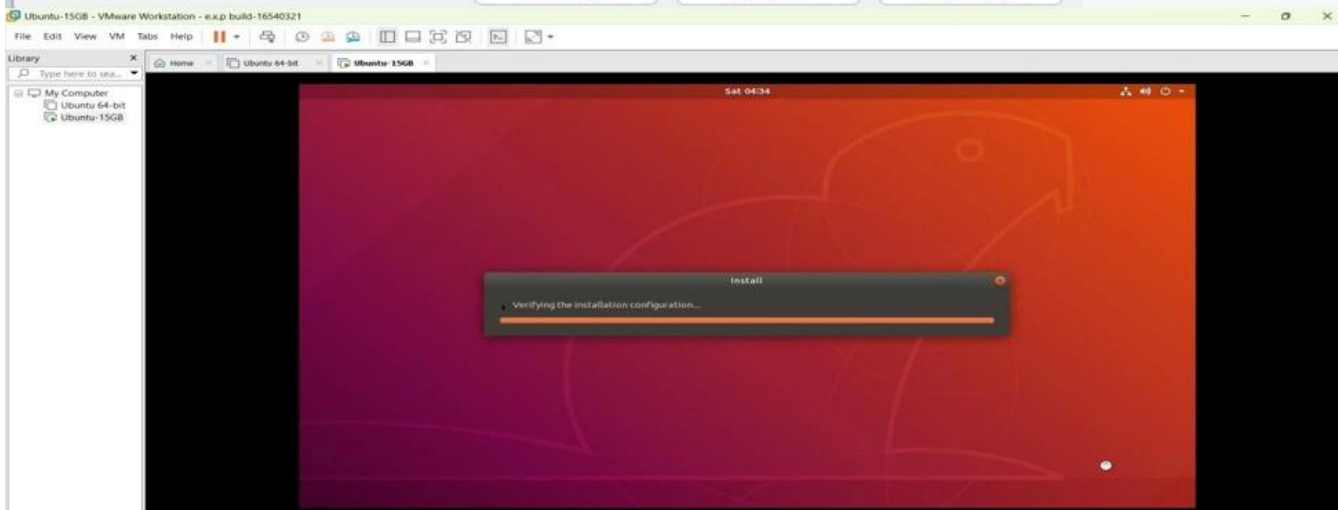
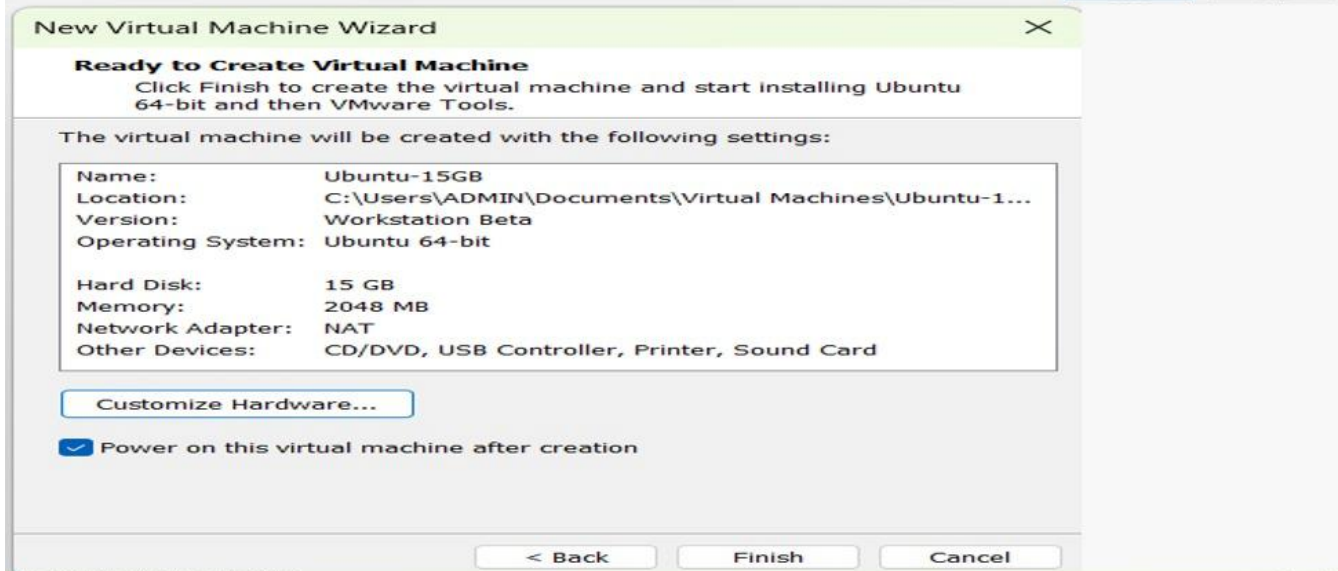
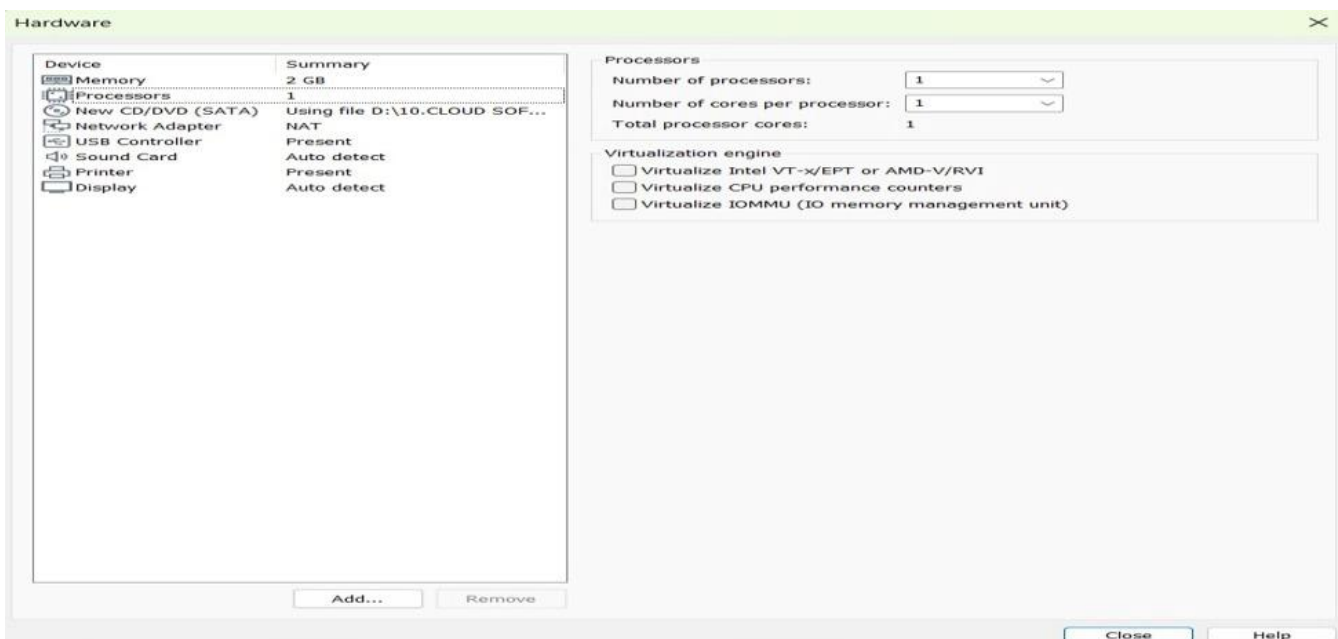
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help

< Back

Next >

Cancel





A.Sai Mrudula

Not listed?



A.Sai Mrudula

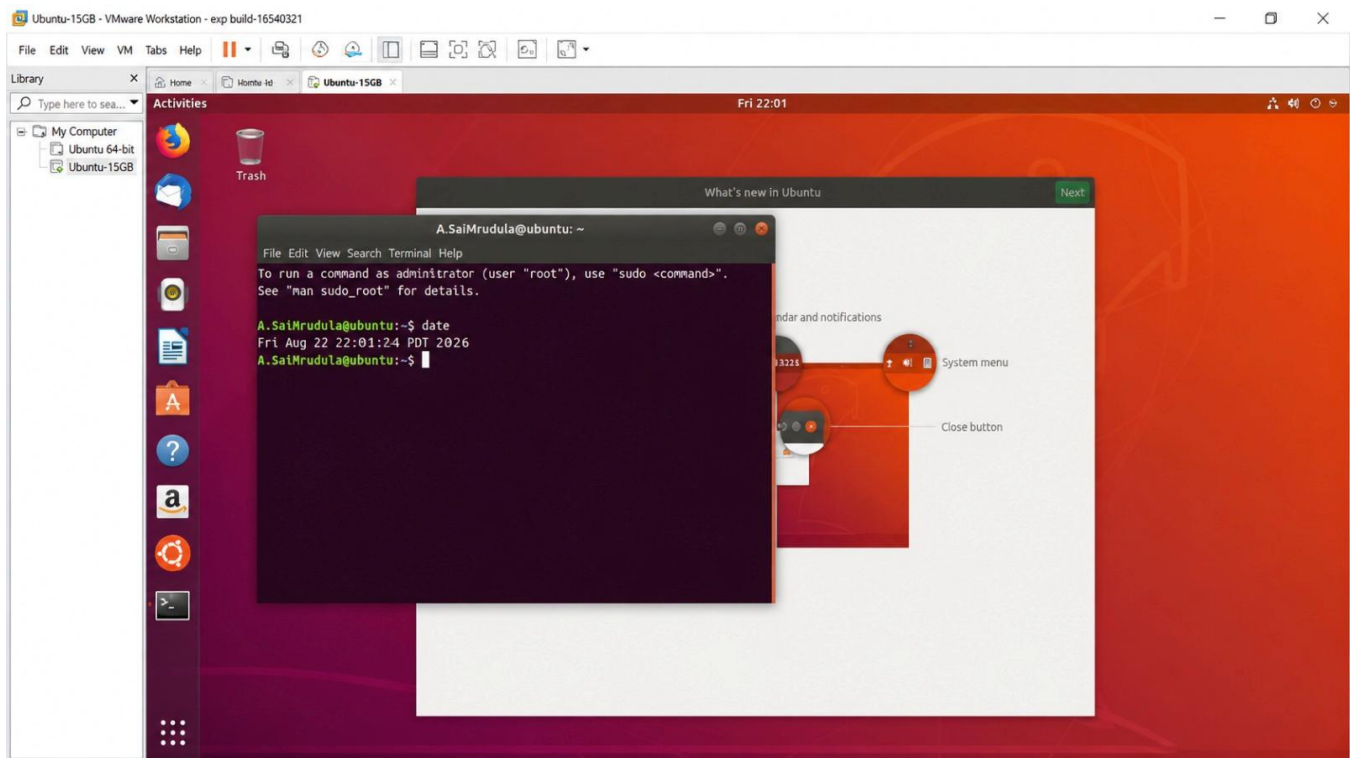
Password:

A dark gray rectangular input field for a password. It contains seven white dots and a white cursor at the end, indicating it is active.

Cancel



Sign In



Result

A Virtual Machine was successfully created using a Type-2 hypervisor with 1 CPU, 2 GB RAM, and 15 GB storage. The guest operating system was installed and the VM was started successfully.